

SolarPro Global — Technical Guide

Engineering basis, standards, and integration reference

Version: 2026-06-05 (beta) **Audience:** Senior engineers, technical leads, and procurement/IT staff at Ghana installer and supplier companies; internal engineering team. **Companion to:** User Guide (workflow) and Sales Pitch (positioning).

This document explains **how** SolarPro computes its outputs and **which standards** it follows, so an external engineer can defend any value the platform produces to a client or to a financing bank.

1. Engineering principles in one page

SolarPro is built on three rules:

1. **Every output is derivable from the inputs.** No "magic" multipliers — every coefficient is listed in this guide or in **assumptions.md**.
2. **Every output is overridable.** Defaults are conservative; the engineer can replace any value with their own.
3. **Every output is traceable.** The proposal PDF shows the calculation path so a reviewing engineer can validate without re-running the tool.

2. Standards applied

Standard	Where it's applied	Notes
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BS 7671 (18th Edition, +A2)	AC cable sizing, derating, protective device coordination	UK origin; default standard for Ghana commercial/industrial wiring
IEC 62548	DC string design, MPPT voltage windows, isolation	International PV-array standard
IEC 60364-7-712	PV-specific installation requirements	Used for protective bonding, isolation
IEC 61730	Module safety classification	Determines Class II isolation requirements
NEC Article 690	Fallback for US-style installations	Triggered only when site country = USA
Ghana Energy Commission — Solar PV Code of Practice (2019)	Permitting checklist on the installation report	Drives the EC-certification section
IEC 61724	Performance ratio & yield calculation	Drives the economic analysis

Standards are cited inline in every report so the bank's engineer can verify.

3. Site assessment

3.1 Irradiance

Source: `config/global_solar_data.py` — region-level peak sun hours (PSH) lookup.

For Ghana:

Region	PSH (kWh/m ² /day)	Ambient (°C)
Greater Accra	5.2	28
Ashanti	5.0	26

Northern	5.6	30
Volta	5.1	27
Western	4.8	26

Source data: NASA POWER + Ghana Meteorological Agency monthly means, averaged 2018–2023.

3.2 ECG tariff

Region-tagged residential and commercial tariffs sourced from Public Utilities Regulatory Commission (PURC) quarterly publications. Escalation default = 6%/year (BoG inflation forecast).

The engineer can override tariff and escalation in the economic step.

4. Load analysis

$$\text{Daily energy demand (Wh/day)} = \sum (P_i \times n_i \times h_i \times d_i / 7)$$

Where:

- P_i = nameplate power of load i (W)
- n_i = quantity
- h_i = hours per day
- d_i = days per week (default 7)

A diversity factor is applied to the **peak** load (not the daily energy):

$$\text{Peak (kW)} = \sum (P_i \times n_i) \times DF$$

Where DF = 0.65 for residential, 0.85 for commercial, 1.0 for industrial. The engineer can change DF per project.

Surge load = peak $\times$ 3 (motor start; configurable 2–5 $\times$).

5. PV array sizing

5.1 Required peak power

$$P_{pv} \text{ (kWp)} = E_{daily} / (PSH \times \eta_{system} \times \delta_{temp})$$

Where:

- E_{daily} = daily energy demand (kWh)
- PSH = peak sun hours from §3.1
- η_{system} = 0.80 (default) — combined inverter + wiring + soiling
- $\delta_{temp} = 1 - \gamma \times (T_{cell} - 25)$ — temperature derate

Cell temperature: $T_{cell} = T_{amb} + 25 \text{ }^{\circ}\text{C}$ (NOCT approximation; configurable). γ (power temperature coefficient) = -0.0034 K^{-1} (Tier-1 mono panel default).

5.2 Module count

$$N_{modules} = \text{ceil}(P_{pv} \times 1000 / W_{module})$$

Where $W_{module} \in \{110, 250, 330, 400, 450, 500, 550\} \text{ Wp}$ (commit [27b2492](#)).

5.3 String design

For each candidate inverter MPPT range $[V_{min}, V_{max}]$:

$$\begin{aligned} N_{series_min} &= \text{ceil}(V_{min} / (V_{mp} \times \delta_{T_hot})) \\ N_{series_max} &= \text{floor}(V_{max} / (V_{oc} \times \delta_{T_cold})) \end{aligned}$$

Temperature corrections:

- $\delta_{T_hot} = 1 + \beta_{voc} \times (T_{cell_hot} - 25) \approx 0.88$ (V_{mp} shrinks when panel is hot)
- $\delta_{T_cold} = 1 + \beta_{voc} \times (T_{cold} - 25) \approx 1.10$ (V_{oc} grows when panel is cold)

$\beta_{voc} = -0.0029 \text{ K}^{-1}$ default.

If `N_series_min > N_series_max`, the MPPT panel is incompatible → engine flags WARNING and suggests a smaller-Vmp module or a different inverter.

5.4 Mounting derate

Mounting type	Soiling factor	Temperature uplift (°C)
<code>rooftop_pitched</code>	0.95	+5
<code>rooftop_flat</code>	0.90	+10
<code>rooftop_metal</code>	0.95	+12
<code>rooftop_membrane</code>	0.95	+8
<code>ground_fixed</code>	0.92	+3
<code>ground_tracking</code>	0.95	+3

These factors fold into `η_system` and `T_cell` automatically.

6. Battery sizing

$$C_{\text{battery}} \text{ (Wh)} = E_{\text{daily}} \times N_{\text{autonomy}} / (\text{DoD} \times \eta_{\text{inv}} \times \eta_{\text{batt}})$$

Defaults:

- `N_autonomy` = 1 day (residential), 0.5 day (grid-tied), 2 days (off-grid critical)
- `DoD` = 0.80 (lithium), 0.50 (gel/AGM)
- `η_inv` = 0.95
- `η_batt` = 0.95 (lithium), 0.85 (lead acid)

The battery type (lithium / gel / AGM) drives DoD and `η_batt`. The engineer chooses; the engine reflects the math.

Output is rounded up to the nearest stocked battery module size (5 kWh, 10 kWh, 15 kWh, 20

kWh by default — configurable in Settings → ROQ)

7. Inverter sizing

$$P_{inv} \text{ (kW)} = \max(\text{Peak_kW} \times 1.25, P_{pv} \times 0.85)$$

Two-sided constraint:

- Must handle peak load with 25% margin
- Must absorb at least 85% of PV array output (Ghana standard avoids over-clipping under low PSH)

A hybrid inverter is selected by default for off-grid + battery; grid-tied selects a string inverter.

8. Cable sizing

8.1 DC cable (panel → combiner → inverter)

Voltage drop method (IEC 62548):

$$A \text{ (mm}^2\text{)} = (2 \times L \times I \times \rho) / \Delta V_{\max}$$

Where:

- L = one-way length (m)
- I = string current at STC
- ρ = 0.0175 $\Omega \cdot \text{mm}^2/\text{m}$ (Cu @ 20 °C)

- $\Delta V_{\max} = 1.5\% \times V_{\text{array}}$

Round up to the next standard size (4, 6, 10, 16, 25 mm²). Verify the result against ampacity at 70 °C ambient — bump up one size if borderline.

8.2 AC cable (inverter → DB → load)

Per BS 7671 §523, Table 4D5 (PVC SWA):

Cable	Current carrying capacity (A) @ 30 °C ambient
2.5 mm ²	24 A
4 mm ²	32 A
6 mm ²	41 A
10 mm ²	57 A
16 mm ²	76 A
25 mm ²	101 A

Derating applied:

- Ambient temperature factor (BS 7671 Table 4B1) — Ghana ambient 32–35 °C → 0.94
- Grouping factor (Table 4C1) — if more than one cable shares conduit
- Installation method factor (clipped direct vs. conduit)

Final: smallest size whose derated capacity $\geq$ design current $\times$ 1.25.

Voltage drop check: $\leq$ 3% for final circuits, $\leq$ 5% combined (per BS 7671 Appendix 12).

8.3 Earthing

Earth conductor sized per BS 7671 §544 — half the phase conductor's cross-section, minimum 4 mm² Cu.

Earth rod: 1.2 m $\times$ 16 mm Cu — measured resistance $\leq$ 10 Ω per IEC 60364-7-712.

9. Bill of Quantities — how it's built

`calc_boq()` traverses the sized system and emits one line per material category:

Category	Quantity logic
Modules	N_modules from §5.2
Mounting rails	ceil(N_modules × 1.05 m / 4.2 m) — 4.2 m standard rail
End clamps	4 × N_strings
Mid clamps	2 × (N_modules - N_strings)
MC4 connectors	2 × N_strings
DC cable	length from §8.1 with 10% wastage
AC cable	length from §8.2 with 10% wastage
Inverter	N_inv
Battery modules	ceil(C_battery / W_battery_unit)
Earth rod	1 per system
Earth conductor	length matches AC cable
Surge arrestor (DC)	1 per MPPT input
Surge arrestor (AC)	1 per phase
Isolators (DC)	1 per string
Combiner box	1 per inverter
Ground mount only	steel post (1 per 4 modules), concrete footing (1 per post), purlin beam (4.2 m per row), earth rod (1 per 20 modules)

Pricing:

- Default: blank — engineer types unit cost.
- With supplier integration enabled: pulls live pricing via the supplier's API. Currently

10 Economic analysis

10.1 Energy yield (per IEC 61724)

$$E_{\text{year}} \text{ (kWh)} = P_{\text{pv}} \times \text{PSH} \times 365 \times \text{PR}$$

PR (performance ratio) defaults:

- Rooftop residential: 0.75
- Rooftop commercial: 0.80
- Ground mount fixed: 0.82
- Ground mount tracking: 0.86

10.2 Cash flow

For year $y \in [1, \text{lifetime}]$:

$$\begin{aligned} \text{Savings}_y &= E_{\text{year}} \times (1 - \delta_y) \times \text{Tariff} \times (1 + \text{escalation})^{(y - 1)} \\ \text{OPEX}_y &= \text{OPEX}_0 \times (1 + 0.03)^{(y - 1)} \\ \text{Net}_y &= \text{Savings}_y - \text{OPEX}_y \\ \text{NPV}_y &= \text{Net}_y / (1 + \text{discount})^y \end{aligned}$$

Where:

- δ_y = degradation per year (default 0.5%, lithium battery system; 0.7% pure PV)
- OPEX_0 = annual maintenance estimate (default 1% of CAPEX)

10.3 Headline metrics

- **Payback (years)** — interpolated to the year where cumulative cashflow ≥ 0
- **IRR (%)** — Newton-Raphson on the cashflow vector, tolerance 1e-6
- **NPV (GHS)** — sum of NPV_y over lifetime
- **LCOE (GHS/kWh)** — $\sum \text{Costs} / \sum \text{Energy}$

11. Reports

Every project produces seven PDF reports, all rendered from the same JSON state:

1. **PV Sizing Report** — array, strings, MPPT match, temperature checks
2. **BOQ** — full materials list with prices
3. **Cable Schedule** — every run with gauge, length, derating, voltage drop
4. **Economic Analysis** — charts + IRR/NPV table
5. **Installation Method Statement** — step-by-step, with safety
6. **Energy Yield Report** — month-by-month projection
7. **Bankable Proposal** — combined cover-page + executive summary + sized excerpts of 1–6

Reports embed the assumptions used (PSH, system efficiency, derating factors) so a third-party engineer can validate without re-running the tool.

12. Data model (for IT staff)

12.1 Storage

Each project's full engineering state lives in a single JSON blob: `projects.data_json`. Schema (abbreviated):

```
{
  "site": { "country": "Ghana", "region": "Greater Accra", "psh": 5.2, "ambient": 30 },
  "loads": [ { "name": "Fridge", "watts": 250, "qty": 1, "hours": 12, "days_per_year": 365 },
  "totals": { "daily_kwh": 18.3, "peak_kw": 4.1, "surge_kw": 12.3 },
  "results": {
    "pv": { "p_kwp": 6.0, "n_modules": 12, "w_module": 500, "strings": [ ... ] },
    "battery": { "kwh": 10, "type": "lithium" },
    "inverter": { "kw": 5.0, "type": "hybrid" },
    "cable": { "dc_mm2": 6, "ac_mm2": 4 },
    "boq": [ ... ],
    "economics": { "payback_y": 4.2, "irr": 0.27, "npv_25y": 86_200 }
  }
}
```

12.2 Export

- **PDF** — **Generate Proposal** button on the project page.
- **JSON** — **GET /api/project/<id>/export.json** (authenticated; returns the schema above).
- **BOQ** — **GET /api/project/<id>/boq.csv** (BOQ only).

12.3 Re-import

POST a previously exported JSON to **/api/project/import** — recreates the project. Useful for restoring from backup or moving a project between accounts.

13. Integration points

For supplier and installer companies, three integration surfaces are available during beta:

13.1 Pricing API (suppliers)

If you sell panels / inverters / batteries / cables, we'll wire your stock into the BOQ procurement screen. Required from you:

- A REST or GraphQL endpoint returning JSON with { **sku**, **name**, **category**, **unit_price_ghs**, **in_stock**, **lead_time_days** }
- Optional webhook for stock updates

We pull on a 6-hour cadence (configurable). Live unit costs replace blank cells in the BOQ.

13.2 Project handoff (installers)

After a designer signs off a project, a **/installer/handoff** webhook fires to your endpoint with the full JSON state. Your project management system picks it up and creates a work order.

13.3 Single sign-on (SSO)

If your company uses Google Workspace, Microsoft 365, or any OIDC provider — we support SSO on Business+ plans. Required: client ID + client secret + issuer URL.

14. Security & compliance

Control	Implementation
Authentication	Username + password; 5-failed-attempt logout 15 min
Session management	HttpOnly Lax cookie; CSRF (<code>_csrf</code>) on every POST
Database isolation	Per-tenant row-level filters on all user data tables
Encryption at rest	Hosting platform-managed (currently AES-256)
Encryption in transit	TLS 1.2+ end-to-end
Audit log	<code>audit.log</code> (JSON-line) — login, project edits, exports
Security headers	CSP, X-Content-Type-Options, Referrer-Policy, Frame-Ancestors
Data export	Self-serve from Settings → Export (per GDPR Art. 20)
Data deletion	Self-serve from Settings → Delete Account
Backups	Daily snapshot, 30-day retention

Reference: `SECURITY.md` in the repo for the full Zero-Trust architecture + RBAC matrix.

15. Performance & SLA targets (beta)

Metric	Target	Notes
Sizing calculation latency	< 30 s	p95
Proposal PDF generation	< 60 s	p95

Optimic	99.9%	Free Railway tier, will rise to 99.9% on RGS
Support response (in-app chat)	< 60 s	Rule-based; AI fallback to Claude/Llama
Support response (email)	24 h	Business hours

16. Where to look in the code (internal staff)

Concern	File
Loads → sizing pipeline	<code>web_app.py</code> → <code>calc_loads</code> / <code>calc_pv</code> / <code>calc_battery</code>
AC cable sizing	<code>calculation/ac_cable_sizing.py</code>
DC cable sizing	<code>web_app.py</code> → <code>size_dc_cable</code>
BOQ generation	<code>web_app.py</code> → <code>calc_boq</code>
Economic engine	<code>web_app.py</code> → <code>calc_economics</code>
Global irradiance data	<code>config/global_solar_data.py</code>
AI helpline	<code>web_app.py</code> → <code>/api/assistant/chat</code> + <code>api_manager.py</code>
Email send chain	<code>api_manager.py</code> → <code>_send_email</code>
Structured logging	<code>logging_config/structured_logger.py</code>
Health endpoints	<code>web_app.py</code> → <code>/api/health*</code>

17. Roadmap (next 90 days)

Month	Theme	Highlights
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June 2026	Data hardening	Custom domain SSL, multi-tenant edge isolation, screenshots in user guide
July 2026	Procurement	Nocheski + Ozo pricing live in BOQ, supplier directory page
2026	Energy reporting	Month-by-month yield, Excel export of BOQ, mobile-optimised dashboard

Feature requests during beta: send to support@aiappinvent.com or via in-app chat. Top-3 by demand each fortnight ships on the next sprint.



18. Screenshots index

The technical guide references the same screens as the user guide:

- `[SCREEN: location-globe]` — for §3 site assessment
- `[SCREEN: loads-input]` — for §4 load analysis
- `[SCREEN: results-pv]` — for §5 PV sizing
- `[SCREEN: boq]` — for §9 BOQ
- `[SCREEN: economics]` — for §10 economics
- `[SCREEN: installation-drawings]` — for §6.4 mounting derate consequences

Capture from the live app, save under `docs/screens/<slug>.png`, rerun the PDF build to embed.

